

❖ Ch1_No.2

If the continuous output power of lasers is 1 W when $\lambda=10 \mu\text{m}$, $\lambda=500 \text{ nm}$, or $\nu=3000 \text{ MHz}$, respectively, what is the number of particles that transit from the upper energy level to the bottom energy level per second?

$$P = \frac{E_{\text{tot}}}{t} \cdot E_0 = h\nu = h \frac{c}{\lambda} \quad \leftarrow \text{in } 1\text{s} \quad (t=1\text{s})$$

$$E_{\text{tot}} = n \cdot E_0 \quad \leftarrow n \text{ is number of particles per second}$$

$$\Rightarrow P = \frac{n \cdot E_0}{t} \quad (t=1\text{s}) = n \cdot E_0 \Rightarrow P = n \cdot E_0 \Rightarrow n = \frac{P}{E_0}$$

we know $P=1\text{W}$, $\lambda_1=10\mu\text{m}$, $\lambda_2=500\text{nm}$, $\nu=3000\text{MHz}$

$$\textcircled{1} \lambda_1=10\mu\text{m} : n = \frac{P}{h \cdot \frac{c}{\lambda}} = \frac{1}{(6.626 \times 10^{-34} \times \frac{3 \times 10^8}{10 \times 10^{-6}})} \approx \underline{5.03 \times 10^{19} \text{ s}^{-1}}$$

$$\textcircled{2} \lambda_2=500\text{nm} : n = \frac{P}{h \cdot \frac{c}{\lambda}} = \frac{1}{6.626 \times 10^{-34} \times \frac{3 \times 10^8}{500 \times 10^{-9}}} \approx \underline{2.52 \times 10^{18} \text{ s}^{-1}}$$

$$\textcircled{3} \nu=3000\text{MHz} : n = \frac{P}{h\nu} = \frac{1}{6.626 \times 10^{-34} \times 3000 \times 10^6} \approx \underline{5.03 \times 10^{23} \text{ s}^{-1}}$$

❖ Ch1_No.3

Suppose the energy level of the laser device is E_2 and E_1 where $f_2=f_1$, the corresponding frequency is ν (wavelength is λ), and the particles number at each energy level is n_2 and n_1 , respectively.

a) When $\nu=3000 \text{ MHz}$, $T=300 \text{ K}$, then $n_2/n_1=?$

b) When $\lambda=1 \mu\text{m}$, $T=300 \text{ K}$, then $n_2/n_1=?$

c) When $\lambda=1 \mu\text{m}$, $n_2/n_1=0.1$, then $T=?$

$$\frac{n_2}{n_1} = \frac{f_2}{f_1} \cdot \exp\left(-\frac{E_2-E_1}{kT}\right), \quad E_0=E_2-E_1=h\nu$$

$$\text{a) } \nu=3000\text{MHz}, T=300\text{K}, f_2=f_1$$

$$\frac{n_2}{n_1} = \exp\left(-\frac{h\nu}{kT}\right) = \exp\left(-\frac{6.626 \times 10^{-34} \times 3000 \times 10^6}{1.38 \times 10^{-23} \times 300}\right) \approx \underline{0.9995}$$

$$\text{b) } \lambda=1\mu\text{m}, T=300\text{K}$$

$$\frac{n_2}{n_1} = \exp\left(-\frac{hc}{kT \cdot \lambda}\right) = \exp\left(-\frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.38 \times 10^{-23} \times 300 \times 10^{-6}}\right) \approx \underline{1.405 \times 10^{-21}}$$

$$c) \lambda = 1 \mu\text{m}, \quad \frac{n_2}{n_1} = 0.1$$

$$\ln \frac{n_2}{n_1} = - \frac{hc}{kT\lambda} \Rightarrow T = - \frac{hc}{k\lambda} \frac{1}{\ln \frac{n_2}{n_1}}$$

$$\Rightarrow T = - \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.38 \times 10^{-23} \times 10^{-6}} \cdot \frac{1}{\ln 0.1} \approx \underline{6.256 \times 10^3 \text{ K}}$$

❖ Ch1_No.6

The transition probability of the spontaneous emission from the energy level of a molecule E_4 to the three lower energy levels E_1 , E_2 , and E_3 is $A_{43}=5 \times 10^7 \text{ s}^{-1}$, $A_{42}=1 \times 10^7 \text{ s}^{-1}$, and $A_{41}=3 \times 10^7 \text{ s}^{-1}$ respectively. Try to find the spontaneous emission lifetime τ_{s4} of E_4 .

If the lifetime of each energy level is $\tau_1=5 \times 10^{-7} \text{ s}$, $\tau_2=6 \times 10^{-9} \text{ s}$, $\tau_3=1 \times 10^{-8} \text{ s}$, and $\tau_4=\tau_{s4}$. when continuously exciting E_4 and reaching the steady-state, try to find the ratio of the number of particles between the corresponding energy levels n_1/n_4 , n_2/n_4 , and n_3/n_4 , and then point out which of the two energy levels has achieved the population inversion.

$$a) A_4 = A_{41} + A_{42} + A_{43} \quad . \quad A_4 = \frac{1}{\tau_{s4}}$$

$$\tau_{s4} = \frac{1}{A_{41} + A_{42} + A_{43}} = \frac{1}{5 \times 10^7 + 1 \times 10^7 + 3 \times 10^7} \approx \underline{1.11 \times 10^{-8} \text{ s}}$$

$$b) \frac{dn_i}{dt} = n_4 A_{4i} - \frac{n_i}{\tau_i} = 0 \Rightarrow \frac{n_i}{n_4} = A_{4i} \cdot \tau_i$$

$$\Rightarrow \frac{n_1}{n_4} = A_{41} \cdot \tau_1 = 3 \times 10^7 \times 5 \times 10^{-7} = \underline{15}$$

$$\Rightarrow \frac{n_2}{n_4} = A_{42} \cdot \tau_2 = 1 \times 10^7 \times 6 \times 10^{-9} = \underline{0.06}$$

$$\Rightarrow \frac{n_3}{n_4} = A_{43} \cdot \tau_3 = 5 \times 10^7 \times 1 \times 10^{-8} = \underline{0.5}$$

c) population inversion condition: $n_{\text{upper}} > n_{\text{lower}}$

$$\Rightarrow \text{we know: } n_3 = 0.5 n_4, \quad n_2 = 0.06 n_4, \quad n_1 = 15 n_4$$

$$\Rightarrow n_1 > n_2, \quad n_1 > n_3, \quad n_1 > n_4$$

$$n_2 < n_3, \quad n_2 < n_4$$

$$n_3 < n_4$$

notice that $n_2 < n_4$, $n_3 < n_4$, $n_2 < n_3$, we could ensure that
 $\Rightarrow$ E_3 and E_2 , E_4 and E_2 , E_4 and E_3 achieve population inversion.